

U-Substitution lets us undo Chain rule

To Undo Product rule we need something new.

Recall $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$

Integrating both sides,

$$f(x)g(x) = \int f'(x)g(x)dx + \int f(x)g'(x)dx$$

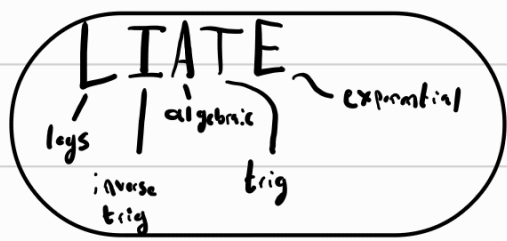
$$\rightarrow \int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$$

let $u = f(x)$, $v = g(x)$, $du = f'(x)$, $dv = g'(x)$

$$\boxed{\int u dv = uv - \int v du}$$

When choosing u , pick functions that get simpler when you derive.

Order to choose $\rightarrow$



Ex: $\int x \sin x \, dx$, $u = x$ $dv = \sin x \, dx$
 $du = dx$ $v = -\cos x$

$$= -x \cos x - \int -\cos x \, dx$$

$$= -x \cos x + \sin x + C$$

Ex: $\int \ln x \, dx$ $u = \ln x$ $dv = dx$
 $du = \frac{1}{x} \, dx$ $v = x$

$$= x \ln x - \int x \cdot \frac{1}{x} \, dx$$

$$= x \ln x - \int dx$$

$$= x \ln x - x$$

Ex: $\int t^2 e^{-t} dt$ $u = t^2$ $dv = e^{-t} dt$
 $du = 2t dt$ $v = -e^{-t}$

$$= -t^2 e^{-t} + 2 \int t e^{-t} dt$$

$u = t$ $dv = e^{-t}$
 $du = dt$ $v = -e^{-t}$

$$= -t^2 e^{-t} + 2 \left[-t e^{-t} + \int e^{-t} dt \right]$$

$$= -t^2 e^{-t} - 2t e^{-t} - 2e^{-t} + C$$

$$\boxed{= -e^{-t} (t^2 + 2t + 2) + C}$$

Ex: $\int e^x \sin x dx$ $u = \sin$, $dv = e^x dx$
 $du = \cos x dx$ $v = e^x$

$$= e^x \sin x - \int e^x \cos x dx, \quad u = \cos x \quad dv = e^x dx$$

$$du = -\sin x dx \quad v = e^x$$

$\int e^x \sin x dx = e^x \sin x - e^x \cos x + \int e^x \sin x dx$

Same

$$2 \int e^x \sin x dx = e^x \sin x - e^x \cos x + C$$

$$\int e^x \sin x dx = \frac{e^x}{2} (\sin x - \cos x) + C$$

Tabular method

Ex:

$$\int x^6 e^{2x} dx$$

$$= + x^6 \frac{1}{2} e^{2x}$$

$$- 6x^5 \frac{1}{4} e^{2x}$$

$$+ 30x^4 \frac{1}{8} e^{2x}$$

$$- 120x^3 \frac{1}{16} e^{2x}$$

$$+ 360x^2 \frac{1}{32} e^{2x}$$

$$- 720x \frac{1}{64} e^{2x}$$

$$+ 720 \frac{1}{128} e^{2x} + C$$

	u	v	Sign
derivate	x^6	integrate e^{2x}	+
	$\downarrow 6x^5$	$\downarrow \frac{1}{2} e^{2x}$	-
	$30x^4$	$\frac{1}{4} e^{2x}$	+
	$120x^3$	$\frac{1}{8} e^{2x}$	-
	$360x^2$	$\frac{1}{16} e^{2x}$	+
	$720x$	$\frac{1}{32} e^{2x}$	-
	720	$\frac{1}{64} e^{2x}$	+
	0	$\frac{1}{128} e^{2x}$	-

multiply

alternates

subtract

add

subtract

step at 0

Definite Integrals

Integration by parts works well with definite integrals as follows

$$\int_a^b f(x)g'(x)dx = f(x)g(x)\Big|_a^b - \int_a^b g(x)f'(x)dx$$

we can evaluate $f(x)g(x)\Big|_a^b$ before or after the entire integration.

Ex:

$$\int_0^1 \arctan x \, dx, \quad \text{let } u = \arctan x, \, dv = dx$$
$$du = \frac{dx}{1+x^2}, \, v = x$$

$$= x \arctan x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} dx$$

$$= (1)\left(\frac{\pi}{4}\right) - 0 - \int_0^1 \frac{x}{1+x^2} dx$$

$$\text{let } u = 1+x^2$$
$$du = 2x dx$$

$$= \frac{\pi}{4} - \frac{1}{2} \int_1^2 \frac{1}{u} du$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln|u| \Big|_1^2 = \boxed{\frac{\pi}{4} - \frac{1}{2} \ln(2)}$$

Reduction formulas

When conducting integration by parts we can end up just reducing the power of a function, this gives us some what helpful formulas called reduction formulas

A well known one is

$$\int \sin^n x dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x dx$$

Pf/

$$\int \sin^n x dx \quad \text{let } u = \sin^{n-1} x \quad dv = \sin x dx$$
$$du = (n-1) \sin^{n-2} x \cos x dx \quad v = -\cos x dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x dx$$

$$\cos^2 x = 1 - \sin^2 x$$

$$= -\sin^{n-1} x \cos x + (n-1) \int (\sin^{n-2} x - \sin^n x) dx$$

$$\int \sin^n x dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx$$

$$n \int \sin^n x dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx$$

$$\int \sin^n x \, dx = -\frac{1}{n} \cos(x) \sin^{n-1}(x) + \frac{n-1}{n} \int \sin^{n-2}(x) \, dx$$